Label Swapping for Imbalanced Music Datasets

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Abstract

Data imbalance is a pervasive issue in machine learning that often hinders classifier performance by biasing predictions towards the majority class. Label-Noise-based Re-balancing (LNR) is a novel method proposed by Hu et al. (2025) that introduces beneficial label noise to adjust biased decision boundaries without the drawbacks of traditional resampling techniques. We apply this method to the imbalanced music dataset Meter2800 for time signature detection, meaning to improve the detection of rare, irregular meters (e.g., 5/4, 7/4) using a ResNet18 architecture. Our results show that beneficial label noise yields slightly improved performance for recognizing minority class time signatures.

1 Introduction

Data imbalance is a common issue in machine learning that hinders classifier performance, leading to biased predictions favoring the majority class. In imbalanced datasets (e.g., 90% class A, 10% class B), accuracy is maximized by favoring class A, pushing the decision boundary deep into the territory of class B.

This results in poor performance for the minority class. Additionally, techniques such as resampling often introduce information loss or overfitting. Finding a solution is relevant in scenarios where over-representation of one class over another may be problematic.

Recently, Hu et al. (2025) proposed a new approach, Label-Noise-based Re-balancing (LNR), suggesting that adding beneficial label noise has the potential to adjust biased decision boundaries and improve performance without the drawbacks of traditional resampling [1].

Our project applies this method to imbalanced music datasets, specifically time signature detection. Meter2800 is an imbalanced dataset of musical excerpts labeled with time signatures, where standard meters 4/4 and 3/4 are abundant, while irregular meters such as 5/4 and 7/4 are scarce [2].

Reproducing the results of ResNet18 on Meter2800 and augmenting the training process with LNR, we seek to establish a new way of detecting complex musical meters.

Code repository is available at `https://github.com/Matdrox/dl_music_label_swapping`

2 Related Work

This project is based on two separate papers. Its purpose is to combine the novel method introduced in the former paper with the architecture of the latter.

2.1 Label Noise Re-balancing (LNR)

Binary classification on imbalanced datasets often results in biased decision boundaries, causing misalignment between accuracy and metrics such as F1-score. LNR addresses this by introducing asymmetric label noise, selectively flipping majority class samples to the minority class based on their similarity to minority class samples. Unlike symmetric noise, which affects all classes equally, LNR focuses on majority samples that are confusing (i.e., they have high posterior probabilities of looking like the minority class). This process pushes the decision boundary back toward the majority region, reclaiming space for the minority class without discarding any data.

2.2 Time Signature Detection

Audio Similarity Matrices (ASM) and Beat Similarity Matrices (BSM) have been used as early methods for meter detection relying on signal processing. More recently, deep learning has shown superior performance. Abimbola et al. (2024) [3] demonstrated that a ResNet18 architecture using Mel-frequency Cepstral Coefficients (MFCCs) as input outperformed traditional machine learning models such as SVM and KNN, as well as other deep neural networks like CNN and CRNN. However, the dataset used in the paper, Meter2800, is highly imbalanced, making it a suitable candidate for testing the limits of the LNR method.

3 Methods

The final model architecture combines ResNet18 for time signature detection with the LNR algorithm for handling data imbalance. Imperfect representation is a classical problem in music information retrieval, where some musical concepts are inherently rare. Time signatures such as 5/4 and 7/4 are uncommon in Western music, while 4/4 and 3/4 dominate. This leads to imbalanced datasets that bias classifiers toward the majority classes, resulting in poor detection of minority class time signatures.

3.1 ResNet18 for Time Signature Detection

Time signature detection is approached as a supervised image classification problem using the methodology instructed by Abimbola et al. (2024) [3]. Raw audio signals are pre-processed into visual representations and fed into a ResNet18 architecture adapted for single-channel audio features.

Audio files are converted into Mel-frequency Cepstral Coefficients (MFCCs), compactly representing the spectral properties of the audio signal. For each 30-second audio clip, 13 cepstral coefficients are extracted over 130 time frames. In order to maintain consistency, the input dimensions are fixed to 130×13 (time frames $\times$ frequency coefficients). Each input sample x_i is thus treated as a single-channel image tensor $x_i \in \mathbb{R}^{1 \times 130 \times 13}$.

The Resnet18 architecture is a deep residual network consisting of 18 layers. While originally designed for three-channel RGB images (e.g., ImageNet), we adapt it for our low-resolution, single-channel audio features. MFCC inputs are processed through a series of residual blocks, letting it learn representations of the musical meters. The final layer is a fully connected layer that maps the extracted features to the output classes corresponding to the time signatures 3/4, 4/4, 5/4, and 7/4.

The original paper implements cross-entropy loss for an Adam optimizer with a learning rate of 0.001 training for 40 epochs. We have decided to work within the LNR framework, which uses SGD with an adaptive learning rate of 0.1 for epochs < 60, 0.001 for epochs < 80 and 0.0001 for epochs ≥ 80. We chose to follow the LNR hyperparameters in order to better study the effects of the method, in a setting that has been optimized for its usage. Additional hyperparameters can be found in Appendix B.

3.2 LNR Algorithm

In order to enforce the model's attention toward the minority class, LNR proposes the concept of beneficial lying. Instead of safely assuming the majority class, flipping the label of ambiguous samples to the minority class causes the model to pay better attention to the nuances of the minority class.

Lying randomly would confuse the model. Therefore, beneficial noise was introduced by the implementation of the LNR algorithm. For the binary case, the noise is created as follows:

- Let X be the input feature space (audio spectograms) and $Y = \{0, 1\}$ be the binary label space, where 0 represents the majority class and 1 the minority class.

- Let f_θ be the ResNet classifier parameterized by weights θ.

- $\hat{\eta}(x_i) = P(Y = 1|X = x_i)$ denotes the estimated posterior probability (confidence score) that sample x_i belongs to the minority class.

- $I_{MA} = \{i|y_i = 0\}$ is the set of indices of majority class samples in the current batch.

The LNR algorithm identifies confusing majority samples based on their posterior probabilities. Calculating the mean μ and standard deviation σ of these probabilities after passing them through the classifier's softmax layer, the standardized Z-score Z_i for each majority sample is computed as:

$$Z_i = \frac{\hat{\eta}(x_i) - \mu}{\sigma}$$

Flip-rate probability $\rho(i)$ is computed using a hyperbolic tangent function and a threshold hyperparameter t_{flip}:

$$\rho(x_i) = \max(\tanh(Z_i - t_{flip}), 0)$$

Labels are flipped stochastically based on $\rho(x_i)$, as only majority samples with sufficiently high confusion (i.e., $Z_i > t_{flip}$) should have a non-zero chance of being flipped to the minority class. The algorithm is given in the pseudocode in the Appendix A.

4 Data

Three datasets are used in this project.

4.1 KEEL

For binary classification experiments replicating the original LNR paper, the same KEEL datasets are used. KEEL is a repository of 99 binary imbalanced datasets. From these, 32 specific datasets were selected by Hu et al. (2025) for their experiments [4].

Their chosen datasets match the following criteria:

a) Baseline already achieves an F1-score over 90%, or all methods have an F1-score lower than 10%.

b) Datasets with insufficient minority class sample size in the test dataset (less than 10).

We picked out four of those selected datasets to replicate the results of: *abalone-17_vs_7-8-9-10*, *ecoli1*, *glass0* and *vehicle1*.

4.2 CIFAR-10

Multiclass reproduction required us to use CIFAR-10, a well-known image classification dataset consisting of 60,000 32x32 color images in 10 classes. CIFAR-100 was also used in the original paper, but it was not necessary for our reproduction.

4.3 Meter2800

Meter2800 is a time signature dataset provided by Harvard Dataverse. It contains four annotated CSV files of audio data segmented into the meter classes 3, 4, 5 and 7. The dataset also contains a total of 2800 audio files of 30 seconds each. It exhibits severe class imbalance, making it a perfect candidate for testing the LNR method. Majority classes 4/4 and 3/4 dominate the dataset with 1200 samples each, while minority classes 5/4 and 7/4 are underrepresented with only 200 samples each.

5 Experiments and Findings

5.1 Binary classification on KEEL datasets

Table 1 shows the results of binary classification. LNR seems to have been the most advantageous for the *abalone-17_vs_7-8-9-10* dataset, and specifically for its F1 score, which might be because of its

Table 1: F1 score, G-mean score, and AUC for CART and MLP classifiers on KEEL datasets

	CART				MLP			
Dataset	Baseline	LNR	SMOTE	CC	Baseline	LNR	SMOTE	CC
	F1 score (CART)				**F1 score (MLP)**			
abalone[...]	0.27±0.09	**0.33±0.07**	0.23±0.05	0.17±0.04	0.21±0.16	**0.37±0.13**	0.33±0.07	0.22±0.05
ecoli1	0.73±0.07	**0.75±0.05**	0.74±0.06	0.71±0.07	0.77±0.06	**0.78±0.06**	0.76±0.06	0.75±0.06
glass0	0.69±0.07	0.69±0.07	**0.70±0.08**	0.66±0.07	0.68±0.11	0.68±0.07	**0.69±0.08**	0.66±0.07
vehicle1	0.50±0.05	0.53±0.05	**0.54±0.05**	0.52±0.05	0.66±0.05	0.69±0.08	0.69±0.04	0.66±0.05
	G-mean score (CART)				**G-mean score (MLP)**			
abalone[...]	0.52±0.10	0.64±0.09	0.68±0.08	**0.74±0.06**	0.33±0.21	0.60±0.18	0.80±0.08	**0.81±0.13**
ecoli1	0.83±0.05	**0.86±0.04**	0.85±0.04	0.84±0.04	0.85±0.05	0.85±0.04	0.87±0.04	0.87±0.04
glass0	0.77±0.05	0.77±0.05	**0.78±0.06**	0.75±0.05	0.76±0.10	**0.77±0.05**	0.76±0.10	0.74±0.06
vehicle1	0.65±0.04	0.68±0.04	0.68±0.04	0.68±0.04	0.77±0.04	**0.81±0.09**	0.80±0.04	0.79±0.04
	AUC (CART)				**AUC (MLP)**			
abalone[...]	0.63±0.05	0.70±0.06	0.71±0.06	**0.75±0.06**	0.91±0.08	0.90±0.08	0.91±0.04	0.90±0.07
ecoli1	0.83±0.05	**0.86±0.04**	0.85±0.04	0.84±0.04	**0.95±0.02**	0.94±0.03	0.94±0.03	0.94±0.03
glass0	0.78±0.05	0.78±0.05	0.78±0.05	0.75±0.05	**0.85±0.06**	0.84±0.05	0.84±0.06	0.82±0.05
vehicle1	0.67±0.03	0.68±0.04	**0.69±0.03**	0.68±0.04	0.89±0.03	0.89±0.05	0.89±0.02	0.87±0.03

relatively big imbalance ratio of 39.31 [**?**]. Otherwise, the results are not clearly in the favor of LNR. The results are similar to the original LNR paper [1] in the sense that LNR outperforms the other methods in the greatest number of scenarios, even though the margins are quite slim.

5.2 Multiclass classification on CIFAR-10

Table 2: Accuracy and ECE for different methods (BRYR VI OSS OM ECE?) (Ett mått på kalibrering?)

Method	Acc@1	Acc@5	H-Acc	M-Acc	T-Acc	ECE
LNR	83.69 ± 0.05	99.01 ± 0.02	86.29 ± 0.10	80.58 ± 0.05	**85.26 ± 0.03**	**9.86 ± 0.07**
MisLAS	**83.73 ± 0.01**	**99.06 ± 0.01**	**86.43 ± 0.03**	**80.63 ± 0.02**	85.14 ± 0.03	9.94 ± 0.01
SelMix	80.55 ± 0.19	96.87 ± 0.19	79.93 ± 0.43	80.12 ± 0.16	81.75 ± 0.35	14.52 ± 1.30

In Table 2 we can see that pure MiSLAS produces the highest overall accuracy, as well as in most subdivisions of data. However, for the tail classes, that is the minority classes with the least examples, LNR comes out on top. This result aligns with LNR's goals of improving minority class performance. By comparing our results with the ones in Table 2 in the original LNR paper [1], there are some observations to be made. Unlike in our results, they achieve the highest overall accuracy when using LNR. Also, SelMix actually produces the best results for their tail classes (few-shot category). One similarity is that pure MiSLAS comes out on top in terms of majority classes.

5.3 Multiclass classification on Meter2800 dataset

After having done a parameter search on the *t_flip* parameter, the confusion matrix on the best setting (*t_flip = 7*) can be seen in Figure 1. We see here that the predicting power of the minority time signatures (5/4 and 7/4) have barely improved. Table 3 gives some insight into why this is. In the 140 epoch setting, both the overall F1 score and the overall accuracy is the highest for *t_flip = 7*. According to the LNR algorithm 1, a higher *t_flip* means that less labels will be flipped on average. Thus, the highest overall accuracies were achieved when the impact of LNR was minimal. However, the tail accuracy (time signatures 5/4 and 7/4) is the highest when *t_flip* is the lowest, as in when the LNR algorithm is the most active.

Table 3: Macro F1-score for different t_{flip} on the Meter2800 validation dataset

t_{flip}	F1 100e	Acc 100e	T-acc 100e	F1 140e	Acc 140e	T-acc 140e
0	0.7643	81.83	**68.0**	0.7792	83.52	**66.83**
1	0.7645	81.86	67.5	0.7787	83.52	66.17
2	**0.7655**	**81.91**	67.5	0.7781	83.50	66.0
3	0.7644	81.88	67.17	0.7788	83.55	66.0
4	0.7644	81.88	67.17	0.7788	83.55	66.0
5	0.7638	81.86	67.0	0.7796	83.60	66.0
6	0.7638	81.86	67.0	0.7796	83.60	66.00
7	0.7638	81.86	67.0	**0.7797**	**83.62**	65.30

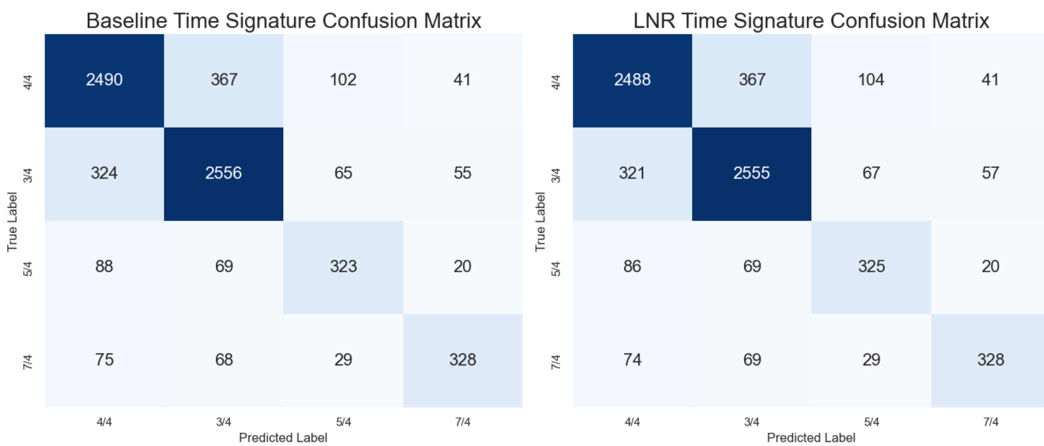

Figure 1: Confusion matrix for LNR on Meter2800 test data with $t_{flip} = 7$ after 140 epochs.

6 Challenges

The LNR paper covers a novel, yet fairly approachable method for imbalanced dataset classification. Algorithms in form of pseudocode are provided, and the method is easy to implement.

However, the paper is quite broad. In an attempt to present the benefits of the proposed LNR method, experiments were conducted using many different frameworks. This requires the reader to not only understand the theory behind the proposed method, but also the details of the various frameworks used in the experiments. This makes it difficult to reproduce the results without prior knowledge of the frameworks.

Additionally, the structure of the paper makes it difficult to follow the experimental setups. The datasets used are scattered throughout the paper, and it is not always clear which datasets were used for which experiments. This requires the reader to constantly refer back to previous (or future) sections to understand the context of the experiments. Even then, some details are left out, making it difficult to fully understand the experimental setups. For example, the implementation of early stopping for multiclass training is neither described in the paper nor made clear in the code.

Upon a close read of the source code provided by the authors of the LNR paper, we found that the original implementation of the CIFAR-10 experiments lacked a distinct validation split and instead uses the test set for early stopping and model selection. A standard validation set is absent in older benchmarks such as the 2009 dataset CIFAR-10, but modern best practices recommend a separate validation set to avoid overfitting to the test set. This may have lead to test data leakage in the original code, if they did use early stopping. In addition, the supplied git repo required a fair amount of work to get running, with several small changes throughout the code needed in order to use the training scripts.

7 Conclusion

The clearest result that we have been able to replicate from the LNR paper [1] is that the method improves minority class performance. In our results, this has also generally been at the cost of general performance across classes, and not just for majority classes.

When applying LNR, it is important to note the impact it might have on overall performance of the model. Even when optimized through parameter search, the overall accuracy isn't guaranteed to outperform other data-level methods. However, if minority classes are of great importance, it might be worth it

8 Ethical Consideration

Time signature detection is a niche task in music information retrieval with limited ethical implications. However, data integrity and representation remain important.

The Meter2800 dataset, while useful for its intended research purpose, may not fully represent the diversity of musical styles and cultures. This could lead to biased models that perform poorly on underrepresented time signatures or genres.

Fixing imbalanced datasets is an important task for machine learning in general. Real-world cases such as breast cancer detection often involve many more negative samples than positive ones, despite the importance of correctly identifying the positive cases. In these critical applications, deliberately mislabeling majority samples to improve sensitivity ought to be done with uttermost care to avoid an unacceptable increase in false positives.

9 Limitations

Meter2800, although imbalanced and suitable for testing LNR, is relatively small with only 200 samples for the minority classes. The statistical significance of the improvement is harder to establish compared to large-scale image datasets like CIFAR-10.

10 Self Assessment

Assessing our project is done using the detailed assessment criteria provided by the course. The final result is a combination of reproduction and novel application of the LNR method to music time signature detection.

10.1 Grades E-D

This paper discusses the LNR method by describing its theory and implementation in detail. We argue why data imbalance is a problem and how LNR addresses it. The paper also reviews related work in the field of imbalanced classification. Coding is done by altering, rewriting, and cleaning up parts of the original LNR paper's GitHub repository.

Both a binary classification using KEEL datasets and a multiclass classification using CIFAR-10 are reproduced successfully. Results closely match those reported in the original paper, confirming a correct reimplementation of the LNR method.

10.2 Grades D-C-B

Grade C requires additional empirical experiments to be documented. This was done by tuning the LNR hyperparameter t_{flip} on the Meter2800 dataset, showing its effect on minority class performance. Meters 5/4 and 7/4 are rare in Western music, making their detection challenging. Observing the resulting macro F1-scores and accuracies for different t_{flip} values, we found that a moderate flip threshold of $t_{flip} = 7$ yielded the best overall accuracy, while $t_{flip} = 0$ gave the best accuracy for the minority class.

10.3 Grade B-A

Going beyond the scope of the original paper, we applied the LNR method to do time signature detection on the imbalanced Meter2800 dataset. This novel application required adapting the LNR algorithm to a different domain and data modality. Audio files were altered to be visualized as MFCC images, suitable for a tailored one-channel ResNet18 input.

Differing thresholds t_{flip} were also tested to find the optimal setting for this task. Doing so helped bring a theoretical method into practical use, along with the team's further examples of real-world applications.

References

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Appendix

A LNR Algorithm Pseudocode

Algorithm 1 Flip labels in binary classification with noise model $\rho(x)$

Require: Feature X and labels Y, Classifier C_f
Require: Threshold t_{flip}
  $Ind_{MA} \leftarrow$ index($Y = 0$)
  $\hat{\eta}, Z, \rho \leftarrow$ zeroVector(size $= N$)
  for $i \in 1 \rightarrow$ length(Y) **do**
    $\hat{\eta}[i] \leftarrow C_f(X[i])$
  end for
  $\mu \leftarrow$ mean($\hat{\eta}[Ind_{MA}]$)
  $\sigma \leftarrow$ std($\hat{\eta}[Ind_{MA}]$)
  for each i_{MA} in Ind_{MA} **do**
    $Z[i_{MA}] \leftarrow \frac{\hat{\eta}[iMA]-\mu}{\sigma}$
    $\rho[i_{MA}] \leftarrow \max(\tanh(Z[i_{MA}] - t_{flip}), 0)$
    $U \leftarrow$ Bernoulli($\rho[i_{MA}]$)
    if $U == 1$ **then**
      $Y[i_{MA}] \leftarrow 1$
    end if
  end for
  return Y

B Hyperparameters

Table 4: Hyperparameters for LNR on Meter2800 Dataset

Hyperparameter	Value
Batch size	128
Initial learning rate	0.1
Learning rate decay schedule	Epochs 60, 80
Momentum	0.9
Weight decay	$2 \cdot 10^{-4}$ (stage 1), $5 \cdot 10^{-5}$ (stage 2)
Number of epochs	100, 140
LNR epoch	80, 100
t_{flip}	0, 1, 2, 3, 4 , 5, 6, 7